Memory Page

▼ FORMULAS

Gauge length(L0)) of a sample given cross sectional area(A0) or circular diameter (d0)

$$L_0 = 5.65\sqrt{A_0}$$

$$L_0 = 5d_0$$

In general, stress is force/original area

$$\text{Engineering Stress} = \frac{Force}{A_0}$$

$$UTS = \frac{F_{max}}{A_0}$$

$$\text{Yield Stress} = \frac{F_{yield}}{A_0}$$

%E and %A

$$\%E = \frac{\Delta L}{L_0} \times 100\% = \frac{L - L_0}{L_0} \times 100\%$$

$$\%A = \frac{\Delta A}{A_0} \times 100\% = \frac{A_0 - A}{A_0} \times 100\%$$

Youngs Modulus

$$E = \frac{Stress}{Strain} = \frac{\frac{F}{A_0}}{\frac{\Delta L,\ extension}{L_0}}$$

▼ STANDARDISED VALUES

Properties & Testing Methods

Hardness Tests

- depth of impression : thickness

  7x for soft, 15x for hard

- Brinells indentor size

  2.5mm, 5mm, 10mm

- F/D^2 for brinells

  10 for soft materials, 30 for harder materials

- Apex angle for Vickers indentor

  136 degrees

- Rockwell Minor and Major loads & indenter
  - HRB

    10kgf, 100 kgf, 1.6mm hardened steel ball indenter

  - HRC

    10kgf, 150 kgf, 120 degree apex diamond cone

  - HRA

    10kgf, 60 kgf, 120 degree apex diamond cone

Plain Carbon Steels

- UCT and LCT for pure iron cooling

  910C and 723C

- Phase change temperatures for pure iron

  1538 → 1400 → 910 → 723

- magnetism change of pure iron

  768C, from para(non) magnetic to ferro(yes)magnetic

- eutectoid, hypo and hyper steel %C
  - eutectoid

    0.8%C

  - hypo-eutectoid

0.022-0.79%C (<0.8%C)

- hyper eutectoid

  0.81%C to 2%C (>0.8%C)

- Steel naming convention, properties & applications
  - Dead mild

    0.05-0.15, soft and ductile, rods chains and car boides

  - Mild

    0.15-0.3, soft and ductile, reinforcement bars ship plate

  - Medium
    - 0.3-0.6 good mix of strenght and toughness, gears and crankshafts
  - high

    0.6-0.8 very strong, rails, wrenches, hammers, screwdrivers

  - carbon tool

    0.8-1.4 hard and brittle, chisells, files, razors

Heat Treatment:

- TABLES

TESTING METHODS	Tensile Test	Brinells (Soft)	Brinells (Hard)	Vickers	Rockwell B	Rockwell C	Rockwell A	Izod/Charpy/Impact test
Load Amount/Type	Tensile Load	F/D^2 ratio of 10	F/D^2 ratio of 30	fixed amount of force	10kgf minor load, 100kgf major	10kgf minor load, 150kgf major	10kgf minor load, 60kgf major	Sudden Impact
Load Applier/Indenter	some pulling thing idk	Hardened Steel ball indenter (2.5mm, 5mm, 10mm)		Square base diamond pyramid (136 deg apex)	1.6mm hardened steel ball	120 deg apex diamond cone (Brale)		Striker
Measured Amount	Force needed to create extension	measured as function of force and diameter		mean of the indentation's diagonals	difference in indentation depth of minor and major load			height difference after striking sample
Sample Type (Metals/Sample prep)	dog bone sample	copper, aluminium	steel/cast iron	very hard materials	softer materials, copper, aluminium, normalised steels	hard materials, hardened steel	hard and thin materials, cemented carbides	notched sample, held both ends with notch away for Charpy, held one end with notch towards for Izod
Advantages		Brinell scale closely relates to tensile strength (3*HBN=UTS)		Accuracy is not dependent on depth of impression (as pyramid makes geometrically similar indents) Useful for measuring very hard materials Negligible surface damage to material due to small indentation.	Rapid in action (as measurement is directly from machine) Suitable for both hard and soft materials (wider range of scale)			
Disadvantages		Very hard materials will deform the indenter		expensive because of diamond more surface prep work needed	too many scales doesnt work for very thin sheets /foils not as reliable as Vickers (in a lab setting) scale does not relate to strength of metal used			

CRYSTAL /MICRO STRUC TURES	Crystal Structures				Microstructures					
	BCC	FCC	HCP/ PCH	BCT	Austenite (γ-iron)	Ferrite (α-iron)	Cementite	Pearlite	Martensite	α-Al
Structure	cubic crystal with 1 atom in center.	Cubic crystal with 1 atom centred at each face	Hexagonal prism crystal with 2 atoms at each pole and 3 in the center	Cuboid crystal with 1 atom in center	BCC solid solution of carbon in iron	FCC solid solution of carbon in iron	Iron carbide in a orthorhombic structure	Alternate lamination of ferrite and cementite	BCT solid solution of carbon in iron	FCC solid solution of aluminium copper compound (Al2Cu) in aluminium
Properties					Soft and ductile, non magnetic	Soft and ductile, magnetic	Hard and brittle	High strength	Very high strength and hardness	Soft and ductile
Examples (if any)	Cr, Fe(ferrite), W	Al, Cu, Ni, Au, Si, Austenite	Mg, Zn, Ti(low temp)	Martensite						

PCS CONVENTION	Carbon%	Properties	Uses	Generalisation
Dead low carbon/Dead mild Steel	0.05-0.15	soft and ductile	car bodies, rods, tubes, chains	For things with complex shapes
low carbon/mild steel	0.15-0.30	soft and ductile	ship plates, reinforcing bars in buildings	For things with complex shapes, however needing strength
medium carbon steel	0.30-0.60	good mix of toughness & hardness	gears, crankshafts	For things under repetitive oscillations/ high fatigue
high carbon steel	0.60-0.80	high strength	railway rails, wrenches, screwdrivers, hammers	for parts with strength needed
carbon tool steel	0.80-1.40	high hardness & wear resistance	chisels, drills, razors, files, axes.	for cutters

HEAT TREAT MENT	Process Annealing	Spheroidising Annealing	Full Annealing	Normalising	Hardening	Tempering (remove stress)	Tempering (add ductility)	Solution Treatment (Al)	Precipitation Hardening (Natural/Artifical Aging)
Purpose	restore ductility, relieve internal stresses for cold worked parts	improve machinability of high carbon/carbon tool steels	Soften steel, form back uniform grain structure	form back uniform grain structure, relieve stress between machining	To improve strength, hardness and wear resistance.	To remove internal stresses due to quenching after hardening	To improve ductility and toughness after hardening	To increase ductility of alloy/ to prep it for precipitation hardening	Increase hardness and strength of aluminium alloy
Steel Type	~0.4%C	>0.5%C	any	any	>0.3%C	hardened steel		Al Alloys	Solution treated Al
Heating Temp	550C to 700C	650C to 700C	30C to 50C above UCT for hypo, 30 to 50C above LCT for hyper (L)	30C to 50C above UCT (V)	30C to 50C above UCT for hypo, 30 to 50C above LCT for hyper (L)	<200C	>200C	550C	Room temp /165C
Holding Time	1h	24h	sufficient/until austenite	sufficient/until austenite	sufficient/until austenite	sufficient		Sufficient/until Cu dissolved (~1h)	4 days/ 10h
Cooling medium	Slow Furnace Cooling			Air cooled	Quenched	Air cooled		Water Quenched	Air cooled
Microstructure	Distorted grains → equiaxed grains	Laminar Pearlite → spheroidised cementite in ferrite lattice	Non-uniform grains → uniform coarse grains	Non-uniform grains → uniform fine grains	Pearlite/Ferrite in steel→ Martensite Cementite uninterrupted	Remains as martensite	Martensite → tempered martensite	coarse precipitates of Cu → supersaturated solid solution of alpha Al	supersaturated solid solution of alpha Al → fine precipitates of Cu
Theory	the heating allows atoms to diffuse in the steel → grains can recrystallise → equiaxed grains replace distorted grains	cementite breaks up into small particles under heating → small spheres of cementite can slide past tool bit/one another → ductile	grains are recrystallised → equiaxed grains are formed —> slower cooling rate forms coarse grains	grains are recrystallised → equiaxed grains are formed —> faster cooling rate forms finer grains	quick cooling rate prevents carbon from diffusing, atoms are displaced slightly →martensite forms, a hard and strong microstructure with BCT crystal structure	heating below 200C does not significantly change microstructure, but relieves internal stresses	heating above 200C reduces the hardness and strength, while increasing toughness and ductility, making the final steel having a good mix of strength and toughness.	Heating to 550C dissolves the precipitates, forming 100% alpha Al, quenching prevents Cu from diffusing out, forming the ductile and soft microstructure	Aging allows the precipitates to form back as finer precipitates, which increases strength

ALLOY STEELS	Shock Tool	Cold Working	Hot Working	High Speed	Ferritic Stainless	Martensitic Stainless	Austenitic Stainless
Composition	S type Steels	O,W or A steels, (oil, water or air cooled)	H type steels	W, M type (W and Mo elements)	14-26% Cr, <0.2%C	12-18%Cr, 0.2-1.2%C	12-24%C, 8-25%Ni, low carbon content.
Advantages	good toughness	good wear resistance/hardness	good hot hardness	good hardness and wear resistance	high ductility thus high malleability and formability,ferromagnetic, can be cold worked, cheapest	strong and hard, can be further hardened or tempered to manipulate properties (same properties as martensite), ferromagnetic	highest corrosion resistance, soft, ductile, biocompatible
Disadvantages					cannot be hardened, lowest corrosion resistance	brittle	non magnetic, cannot be heat treatedsusceptible to intergranular corrosion
Applications	for tools undergoing repetitive impact, e.g. chisels	for tools used during cold working, e.g. punches	for tools used during hot working, e.g. mould cavities, dies, etc.	for tools that machine at high speeds, e.g. drill bits, saws	trims, utensils deep drawn articles and kitchen sinks (complex shapes/moulded by cold working)	cutting tools, knives, blades, scissors, surgical blades.	food handling items and surgical implants
Standard Examples	S7	O1, W1, A2.	H12	T1, M2, 18/4/1, 18%W, 4%Cr, 1%V	430	410	18/8 (18%Cr, 8%Ni), or 302

ALLOY DESIGNATION	Designation	Composition	Properties (if any)		Applications (if any)
Alloy Steel	10xx	C (plain carbon steel)			
	23xx	Ni			
	31xx	Ni Cr			
	43xx	Ni Cr Mo			
Wrought Aluminium	1xxx	Al (~pure)	generally ductile and malleable Cu Mn Si Mg (Si Mg) Zn	Non heat treatable	
	2xxx	Al Cu		Heat treatable	aircrafts
	3xxx	Al Mn		Non heat treatable	corrugated roofs, window frames
	4xxx	Al Si		Nil	forged piston
	5xxx	Al Mg		Non heat treatable	train/bus bodies
	6xxx	Al Mg Si		Heat treatable	Extruded profiles, scaffolds,
	7xxx	Al Zn		Heat treatable	aircrafts
Cast Aluminium	2xx.x	Al Cu	Has good fluidity and low freezing range		piston head
	4xx.x	Al Si			oil slumps, gear boxes, radiators.

CASTING METHODS	Green Sand Casting	CO2 Casting	Investment Casting	Hot Chamber Casting	Cold Chamber Casting
Cast Material	High m.p. metals (cast iron, Al)	Sodium Silicate solution	Unlimited choice of metals	Low m.p. metals (Pb, Zn, Sn)	Low m.p. metals (Al, Cu, Mg)
Mould	Silica sand, water, Bentonite Clay, in 2 parts, cope on drag	Sand + Sodium Silicate with CO2 gas to	Ceramic slurry/ investment slurry	Alloy Steel	
Process	Metal is poured into the pouring basin and sprue, through the runner and up the riser and into the mould cavity through the gate(s).	Sand and Sodium Silicate is added to the mould, and is rammed before CO2 gas is added to form a hard silica gel. The mould is then knocked out.	Pattern is injection moulded and assembled. Ceramic shell is made(investing & stuccoing) and then dewaxed at 300C and fired at 1000C to obtain the mould. metal is poured into the mould (casting) and the ceramic shell is knocked out after cooled	A plunger forces the molten metal in the shot chamber to flow into the die and cool inside it. Ejector pins open the die to remove the cast.	Molten metal is poured from the pouring aperture to the cylindrical chamber, where a piston forces the molten metal into the die, before being cooled and ejected.
Result/Cast	Cast has poor surface finishing, low dimensional accuracy	Cast has good surface finish, low dimensional accuracy.	Cast has very good surface finishing, good dimensional accuracy.	Cast has very good surface finishing, very good dimensional accuracy	Cast has very good surface finishing, very good dimensional accuracy
Applications	Machine beds	Cores, Moulds	large components like turbine blades, or complex shapes like jewelry and implants	Al Gear boxes, oil sumps,steam iron soleplates, zinc toy cars (anything that may need to be produced at large scale)	
Advantages	Cheap (at small scale), wide range of alloys, works well with high m.p. metals, can cast complex moulds,	Fast production rate, produces strong cores without baking, less need for taper, cheap as well	Very wide range of metals, good surface finish and dimensional accuracy, works well with large or intricate parts.	Very fast production rate, good surface finishing and dimensional accuracy	Prolonged immersion is avoided, high injection pressure reduces pores and shrinkage in the cast.
Disadvantages	poor surface finish, low dimensional accuracy	sand used cannot be recycled + similar disadvantages of sand casting + hard to knockout mould after casting	Process has slow production rate and is expensive	Limited to low m.p. alloys, does not work with Al due to iron contamination, high risk of trapping air in the cast. Equipment is also expensive	Some porosity is still possible due to alloy steel mould